Finite Element Method

Approximations of Trial Solutions for 2D and 3D Problems — Part 2

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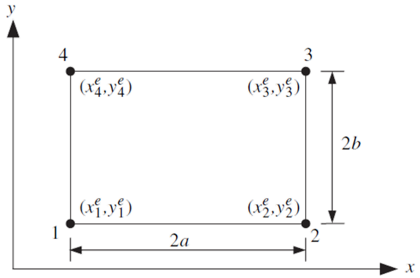
- We first consider a **four-node rectangular element** as depicted in the figure.
- The two-dimensional shape functions for a rectangular element are obtained as a **product of the one-dimensional shape functions** as illustrated in the figure.



Four-node rectangular element.

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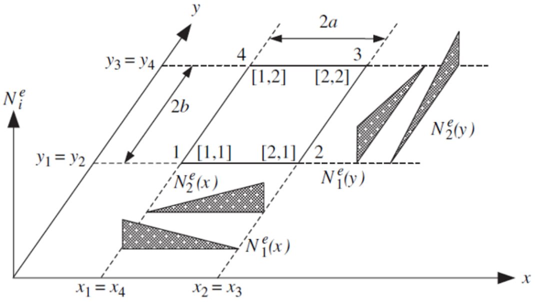


Figure 7.12 Construction of two-dimensional shape functions.

Construction of two-dimensional shape functions.

- The two-dimensional shape function can be written as a product of the one-dimensional shape functions by

$$N^e_{[I,J]}(x,y) = N^e_I(x)\,N^e_J(y) \quad \text{for } I=1,2 \text{ and } J=1,2. \tag{1}$$

- The above two-dimensional shape function has the **Kronecker delta property**:

$$N^e_{[I,J]}(x^e_M, y^e_L) = N^e_I(x_M)\,N^e_J(y_L) = \delta_{IM}\,\delta_{JL}. \tag{2}$$

Shape functions of the four-node rectangle (last column) constructed from one-dimensional shape functions (nodal values in columns 4–7).

K	I	J	$N^e_1(x^e_I)$	$N^e_2(x^e_I)$	$N^e_1(y^e_I)$	$N^e_2(y^e_I)$	2D: $N^e_K(x,y) = N^e_{[I,J]}(x,y)$
1	1	1	1	0	1	0	$N^e_1(x)\,N^e_1(y)$
2	2	1	0	1	1	0	$N^e_2(x)\,N^e_1(y)$
3	2	2	0	1	0	1	$N^e_2(x)\,N^e_2(y)$
4	1	2	1	0	0	1	$N^e_1(x)\,N^e_2(y)$

- From the table and Eq. (2), the two-dimensional shape functions are

$$
\begin{aligned}
N_1^e(x,y) &= \frac{x - x_2^e}{x_1^e - x_2^e} \cdot \frac{y - y_4^e}{y_1^e - y_4^e} = \frac{1}{A^e}(x - x_2^e)(y - y_4^e), \\
N_2^e(x,y) &= \frac{x - x_1^e}{x_2^e - x_1^e} \cdot \frac{y - y_4^e}{y_1^e - y_4^e} = -\frac{1}{A^e}(x - x_1^e)(y - y_4^e), \\
N_3^e(x,y) &= \frac{x - x_1^e}{x_2^e - x_1^e} \cdot \frac{y - y_1^e}{y_4^e - y_1^e} = \frac{1}{A^e}(x - x_1^e)(y - y_1^e), \\
N_4^e(x,y) &= \frac{x - x_2^e}{x_1^e - x_2^e} \cdot \frac{y - y_1^e}{y_4^e - y_1^e} = -\frac{1}{A^e}(x - x_2^e)(y - y_1^e),
\end{aligned}
\tag{3}
$$

where A^e is the area of the element. The element shape functions are shown in the figure.

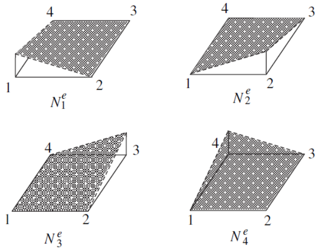


Figure 7.13 Graphical illustration of rectangular element shape functions.

Graphical illustration of rectangular element shape functions.

- The shape functions in Eq. (3) are **only suitable for rectangular elements**.
- To treat a greater variety of four-node quadrilateral shapes shown in Figure 7.14, a more powerful method must be developed.
- The shape functions in Eq. (2) are *not linear* along the edges of an arbitrary quadrilateral element, so two common nodes do not suffice to ensure C^0 continuity between elements.
- Resolving this quandary has led to one of the most important developments in finite elements: the **isoparametric element**.
- The isoparametric concept enables one to construct elements with **curved sides**, which are very powerful in modelling many complex engineering structures.

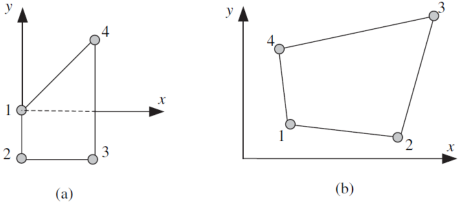


Figure 7.14 Four-node quadrilateral elements.

Figure 7.14 Four-node quadrilateral elements.

- Recall that we defined a standard domain $[-1, 1]$ and then mapped that standard domain into the physical domain of the finite element by

$$x = x_1^e N_1^e(\xi) + x_2^e N_2^e(\xi) = x_1^e \, \frac{1-\xi}{2} + x_2^e \, \frac{1+\xi}{2}, \qquad \xi \in [-1, 1]. \tag{4}$$

  We call the domain $[-1, 1]$ the *parent element domain* and ξ the *parent coordinate* (also called a *natural coordinate*).

- Starting with the shape-function expression for the field and substituting in (4), we obtain

$$\begin{aligned} \theta &= \theta_1^e \, \frac{x - x_2^e}{x_1^e - x_2^e} + \theta_2^e \, \frac{x - x_1^e}{x_2^e - x_1^e} \\ &= \theta_1^e \, \frac{x_1^e(1-\xi) + x_2^e(1+\xi) - 2x_2^e}{2(x_1^e - x_2^e)} + \theta_2^e \, \frac{x_1^e(1-\xi) + x_2^e(1+\xi) - 2x_1^e}{2(x_2^e - x_1^e)} = \theta_1^e \, \frac{1-\xi}{2} + \theta_2^e \, \frac{1+\xi}{2}. \end{aligned} \tag{5}$$

- The form of the linear approximation is **identical** to the map from the parent element to the physical element.

- To develop a quadrilateral element, we let the parent element be a **biunit square** as shown in the figure.
- We map the physical element from the parent element by the four-node shape functions:

$$x(\xi,\eta) = \mathbf{N}^{4\mathrm{Q}}(\xi,\eta)\,\mathbf{x}^e, \qquad y(\xi,\eta) = \mathbf{N}^{4\mathrm{Q}}(\xi,\eta)\,\mathbf{y}^e \tag{6}$$

where $\mathbf{x}^e = [x_1^e\ x_2^e\ x_3^e\ x_4^e]^{\mathrm{T}}$, $\mathbf{y}^e = [y_1^e\ y_2^e\ y_3^e\ y_4^e]^{\mathrm{T}}$.

- The notation $\mathbf{N}^{4\mathrm{Q}}$ emphasises that the shape functions are **identical for every quadrilateral element** (independent of element coordinates).

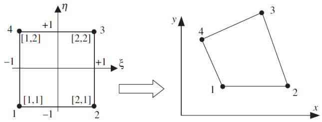


Figure 7.15 Mapping from the parent to the physical Cartesian coordinate system; brackets enclose the dyadic node numbers for the tensor product approach to the construction of two-dimensional shape functions from one-dimensional shape functions.

Mapping from the parent to the physical Cartesian coordinate system; brackets enclose the dyadic node numbers.

- The shape functions are obtained by replacing (x, y) with (ξ, η) and the nodal coordinates (x_I, y_I) with (ξ_I, η_I) in the parent element. Using Eq. (3), the resulting shape functions are:

$$N_I^{4\mathrm{Q}}(\xi, \eta) = \tfrac{1}{4}(1 + \xi_I\,\xi)(1 + \eta_I\,\eta) \tag{7}$$

where (ξ_I, η_I) are nodal coordinates in the parent element:

Node I	ξ_I	η_I
1	-1	-1
2	$+1$	-1
3	$+1$	$+1$
4	-1	$+1$

Nodal coordinates in the parametric element domain.

- The trial solution is approximated by the same shape functions:

$$\theta^e(\xi, \eta) = \mathbf{N}^{4\mathrm{Q}}(\xi, \eta)\,\mathbf{d}^e \tag{8}$$

- In terms of the physical coordinates, the gradient of the trial solution is

$$\nabla\theta^e = \mathbf{B}^e\,\mathbf{d}^e, \tag{9}$$

where

$$\mathbf{B}^e = \begin{bmatrix} \dfrac{\partial N_1^{4Q}}{\partial x} & \dfrac{\partial N_2^{4Q}}{\partial x} & \dfrac{\partial N_3^{4Q}}{\partial x} & \dfrac{\partial N_4^{4Q}}{\partial x} \\[2ex] \dfrac{\partial N_1^{4Q}}{\partial y} & \dfrac{\partial N_2^{4Q}}{\partial y} & \dfrac{\partial N_3^{4Q}}{\partial y} & \dfrac{\partial N_4^{4Q}}{\partial y} \end{bmatrix}. \tag{10}$$

- To obtain the derivatives of the shape functions with respect to the physical coordinates (x, y), we apply the **chain rule**:

$$\frac{\partial N_I^{4Q}}{\partial \xi} = \frac{\partial N_I^{4Q}}{\partial x}\frac{\partial x}{\partial \xi} + \frac{\partial N_I^{4Q}}{\partial y}\frac{\partial y}{\partial \xi}, \qquad \frac{\partial N_I^{4Q}}{\partial \eta} = \frac{\partial N_I^{4Q}}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial N_I^{4Q}}{\partial y}\frac{\partial y}{\partial \eta}, \tag{11}$$

or in matrix form:

$$\begin{bmatrix} \dfrac{\partial N_I^{4Q}}{\partial \xi} \\[2ex] \dfrac{\partial N_I^{4Q}}{\partial \eta} \end{bmatrix} = \underbrace{\begin{bmatrix} \dfrac{\partial x}{\partial \xi} & \dfrac{\partial y}{\partial \xi} \\[2ex] \dfrac{\partial x}{\partial \eta} & \dfrac{\partial y}{\partial \eta} \end{bmatrix}}_{\mathbf{J}^e} \begin{bmatrix} \dfrac{\partial N_I^{4Q}}{\partial x} \\[2ex] \dfrac{\partial N_I^{4Q}}{\partial y} \end{bmatrix}. \tag{11}$$

The matrix $\mathbf{J}^e$ is the **Jacobian matrix**. The required derivatives are obtained by inverting this expression.

- In concise matrix form:

$$\begin{bmatrix} \frac{\partial N_I^{4\mathrm{Q}}}{\partial x} \\ \frac{\partial N_I^{4\mathrm{Q}}}{\partial y} \end{bmatrix} = (\mathbf{J}^e)^{-1} \begin{bmatrix} \frac{\partial N_I^{4\mathrm{Q}}}{\partial \xi} \\ \frac{\partial N_I^{4\mathrm{Q}}}{\partial \eta} \end{bmatrix}, \qquad \mathbf{J}^e = \begin{bmatrix} \partial x/\partial \xi & \partial y/\partial \xi \\ \partial x/\partial \eta & \partial y/\partial \eta \end{bmatrix}, \tag{12}$$

$$\nabla N_I^{4\mathrm{Q}} = (\mathbf{J}^e)^{-1} \mathbf{G} N_I^{4\mathrm{Q}} \tag{13}$$

where $\mathbf{G}$ is the gradient operator in the parent coordinate system:

$$\mathbf{G} = \begin{bmatrix} \partial/\partial \xi \\ \partial/\partial \eta \end{bmatrix}. \tag{14}$$

- The Jacobian matrix can be written explicitly as

$$\mathbf{J}^e = \begin{bmatrix} \frac{\partial N_1^{4\mathrm{Q}}}{\partial \xi} & \frac{\partial N_2^{4\mathrm{Q}}}{\partial \xi} & \frac{\partial N_3^{4\mathrm{Q}}}{\partial \xi} & \frac{\partial N_4^{4\mathrm{Q}}}{\partial \xi} \\ \frac{\partial N_1^{4\mathrm{Q}}}{\partial \eta} & \frac{\partial N_2^{4\mathrm{Q}}}{\partial \eta} & \frac{\partial N_3^{4\mathrm{Q}}}{\partial \eta} & \frac{\partial N_4^{4\mathrm{Q}}}{\partial \eta} \end{bmatrix} \begin{bmatrix} x_1^e & y_1^e \\ x_2^e & y_2^e \\ x_3^e & y_3^e \\ x_4^e & y_4^e \end{bmatrix} \tag{15}$$

or in compact form:

$$\mathbf{J}^e = \mathbf{G}\mathbf{N}^{4\mathrm{Q}} \left[\mathbf{x}^e \quad \mathbf{y}^e\right]. \tag{16}$$

- Using Eqs. (10), (15), and (16), we can write

$$\mathbf{B}^e = (\mathbf{J}^e)^{-1}\mathbf{G}\mathbf{N}^{4\mathrm{Q}}. \tag{17}$$

- For the mapping (7) to be **unique** at each point, it is necessary that the determinant of the Jacobian be nonzero. Furthermore, the determinant must be *positive*, so we require

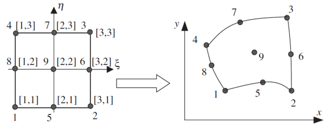
$$|\mathbf{J}^e| \equiv \det(\mathbf{J}^e) > 0 \qquad \forall\, e \text{ and } (x, y). \tag{18}$$

- This requirement is fulfilled if all interior angles in all quadrilaterals are less than $180°$.

Note on superscripts

Although the shape functions $\mathbf{N}^{4\mathrm{Q}}$ do not depend on element coordinates, the Jacobian matrix $\mathbf{J}^e$ and the derivative matrix $\mathbf{B}^e$ depend on the element coordinates as seen from Eqs. (16) and (17). Therefore, the superscript appearing in the isoparametric shape functions denotes the *element type*, whereas in $\mathbf{J}^e$ and $\mathbf{B}^e$ it denotes the *element number*.

- Higher-order isoparametric elements provide one of the most attractive features of finite elements: the ability to **model curved boundaries**.
- As an example of a curved-sided isoparametric element, we describe the **nine-node quadratic element**. The parent and physical element domains are shown in the figure.
- The **node numbering convention** is as follows. The corner nodes are numbered first, followed by the mid-side nodes, both in the counterclockwise direction; the first mid-side node is defined between nodes 1 and 2, and the internal node is numbered last.

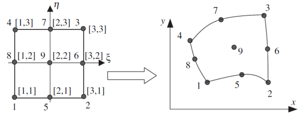


Nine-node isoparametric quadrilateral in parent and physical domains; brackets enclose the dyadic node numbers for the tensor product approach to the construction of two-dimensional shape functions from one-dimensional shape functions.

Nine-node isoparametric quadrilateral in parent and physical domains; brackets enclose the dyadic node numbers.

- To generate the shape functions for the nine-node quadrilateral by the **tensor product method**, we take the product of the three-node shape functions in terms of ξ with the three-node shape functions in terms of η:

$$N_K^{9Q}(\xi,\eta) = N_{[I,J]}^{9Q}(\xi,\eta) = N_I^{3L}(\xi)\, N_J^{3L}(\eta), \quad -1 \le \xi, \eta \le 1. \tag{19}$$



Nine-node isoparametric quadrilateral in parent and physical domains; brackets enclose the dyadic node numbers for the tensor product approach to the construction of two-dimensional shape functions from one-dimensional shape functions.

Nine-node isoparametric quadrilateral (parent domain with dyadic labels).

Node locations in natural coordinates:

- N_1: $(-1,-1)$ — bottom-left corner
- N_2: $(+1,-1)$ — bottom-right corner
- N_3: $(+1,+1)$ — top-right corner
- N_4: $(-1,+1)$ — top-left corner
- N_5: $(0,-1)$ — midpoint of bottom edge
- N_6: $(+1,0)$ — midpoint of right edge
- N_7: $(0,+1)$ — midpoint of top edge
- N_8: $(-1,0)$ — midpoint of left edge
- N_9: $(0,0)$ — centre of element

- The shape functions for the nine-node quadrilateral element are constructed as products of linear and quadratic terms in ξ and η:

Corner nodes:

$$\begin{aligned} N_1 &= \tfrac{1}{4}(1-\xi)(1-\eta)(\xi+\eta-1), \\ N_2 &= \tfrac{1}{4}(1+\xi)(1-\eta)(\xi-\eta-1), \\ N_3 &= \tfrac{1}{4}(1+\xi)(1+\eta)(\xi+\eta+1), \\ N_4 &= \tfrac{1}{4}(1-\xi)(1+\eta)(\xi-\eta+1). \end{aligned}$$

Edge midpoints:

$$\begin{aligned} N_5 &= \tfrac{1}{2}(1-\xi^2)(1-\eta), \\ N_6 &= \tfrac{1}{2}(1+\xi)(1-\eta^2), \\ N_7 &= \tfrac{1}{2}(1-\xi^2)(1+\eta), \\ N_8 &= \tfrac{1}{2}(1-\xi)(1-\eta^2). \end{aligned}$$

Centre node:

$$N_9 = (1-\xi^2)(1-\eta^2).$$

Partition of Unity

The sum of all shape functions is equal to 1 at any point within the element:

$$\sum_{i=1}^{9} N_i = 1.$$

- In an isoparametric element, the approximation and the map from the parent to the physical plane are generated by the **same shape functions**. Thus, for the nine-node quadrilateral:

$$x(\xi,\eta) = \mathbf{N}^{9\mathrm{Q}}(\xi,\eta)\,\mathbf{x}^e, \quad y(\xi,\eta) = \mathbf{N}^{9\mathrm{Q}}(\xi,\eta)\,\mathbf{y}^e, \quad \theta^e(\xi,\eta) = \mathbf{N}^{9\mathrm{Q}}(\xi,\eta)\,\mathbf{d}^e. \tag{20}$$

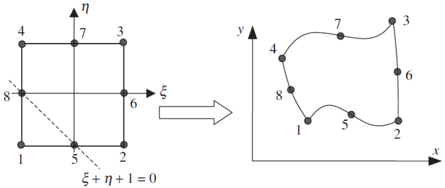
- An important feature of this element is that the **edges are curved** in the physical domain.
- The $\mathbf{B}^e$ matrix is obtained by the same procedure as before:

$$\mathbf{B}^e = \begin{bmatrix} \dfrac{\partial N_1^{9\mathrm{Q}}}{\partial x} & \dfrac{\partial N_2^{9\mathrm{Q}}}{\partial x} & \cdots & \dfrac{\partial N_9^{9\mathrm{Q}}}{\partial x} \\ \dfrac{\partial N_1^{9\mathrm{Q}}}{\partial y} & \dfrac{\partial N_2^{9\mathrm{Q}}}{\partial y} & \cdots & \dfrac{\partial N_9^{9\mathrm{Q}}}{\partial y} \end{bmatrix} \tag{21}$$

or equivalently $\mathbf{B}^e = (\mathbf{J}^e)^{-1}\mathbf{G}\mathbf{N}^{9\mathrm{Q}}$, where

$$\mathbf{J}^e = \begin{bmatrix} \dfrac{\partial N_1^{9\mathrm{Q}}}{\partial \xi} & \cdots & \dfrac{\partial N_9^{9\mathrm{Q}}}{\partial \xi} \\ \dfrac{\partial N_1^{9\mathrm{Q}}}{\partial \eta} & \cdots & \dfrac{\partial N_9^{9\mathrm{Q}}}{\partial \eta} \end{bmatrix} \begin{bmatrix} x_1^e & y_1^e \\ \vdots & \vdots \\ x_9^e & y_9^e \end{bmatrix}, \qquad \mathbf{G}\mathbf{N}^{9\mathrm{Q}} = \begin{bmatrix} \partial/\partial\xi \\ \partial/\partial\eta \end{bmatrix} \begin{bmatrix} N_1^{9\mathrm{Q}} & \cdots & N_9^{9\mathrm{Q}} \end{bmatrix}. \tag{22, 23}$$

- Commercial software usually employs the formulation of higher-order elements **without internal nodes**, as shown in the figure. These are called **serendipity elements**.



Eight-node serendipity element in parent and physical domains.

Node locations in natural coordinates:

- N_1: $(-1,-1)$; N_2: $(+1,-1)$; N_3: $(+1,+1)$
- N_4: $(-1,+1)$; N_5: $(0,-1)$; N_6: $(+1,0)$
- N_7: $(0,+1)$; N_8: $(-1,0)$

Corner nodes:

$$
\begin{aligned}
N_1 &= \tfrac{1}{4}(1-\xi)(1-\eta)(-1-\xi-\eta),\\
N_2 &= \tfrac{1}{4}(1+\xi)(1-\eta)(-1+\xi-\eta),\\
N_3 &= \tfrac{1}{4}(1+\xi)(1+\eta)(-1+\xi+\eta),\\
N_4 &= \tfrac{1}{4}(1-\xi)(1+\eta)(-1-\xi+\eta).
\end{aligned}
$$

Midpoint nodes:

$$
\begin{aligned}
N_5 &= \tfrac{1}{2}(1-\xi^2)(1-\eta),\\
N_6 &= \tfrac{1}{2}(1+\xi)(1-\eta^2),\\
N_7 &= \tfrac{1}{2}(1-\xi^2)(1+\eta),\\
N_8 &= \tfrac{1}{2}(1-\xi)(1-\eta^2).
\end{aligned}
$$

Properties of the Shape Functions

1. **Partition of Unity**: The sum of all shape functions is equal to 1 at any point inside the element:

$$\sum_{i=1}^{8} N_i = 1.$$

2. **Nodal Interpolation**: Each shape function N_i equals 1 at its corresponding node and 0 at all other nodes.
3. **Continuity**: The shape functions ensure C^0 continuity across element boundaries.

Applications

The 8-node serendipity element is widely used for:

- 2D problems in elasticity.
- Heat transfer and other field problems where mid-side nodes provide better accuracy than a 4-node quadrilateral.

Next: Approximations of Trial Solutions for 2D and 3D Problems — Part 3

Thanks for your attention!